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SPARTIFICIAL

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GR4VI7Y_SPY: Exploring Fusion Strategies for Multiview Classification of Glitches in Gravitational Wave Strain Data.

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Abstract

Gravitational waves, ripples in spacetime caused by the acceleration of massive objects, represent one of the most profound predictions of Albert Einstein's general theory of relativity, formulated in 1916.^[1] These waves propagate outward at the speed of light, carrying information about their cataclysmic origins, such as binary star systems and black hole mergers. The first direct detection of these waves occurred in 2015, a century after Einstein's prediction, when the LIGO (Laser Interferometer Gravitational-Wave Observatory) detectors in Livingston, Louisiana, and Hanford, Washington, captured the signature of a black hole merger.^[2] This groundbreaking observation confirmed the existence of gravitational waves and opened a new window for astrophysical exploration.

LIGO's extraordinary sensitivity allows it to detect the minute distortions in spacetime caused by some of the most energetic events in the universe. However, this sensitivity also makes LIGO susceptible to various forms of non-cosmological noise, or "glitches", which can interfere with the detection of genuine gravitational wave signals. These glitches can arise from a multitude of sources, ranging from environmental disturbances to instrumental artifacts, posing a significant challenge for the accurate identification and analysis of gravitational waves.

To address this issue, we propose a machine learning application aimed at classifying and mitigating these glitches. By utilizing advanced machine learning techniques, particularly convolutional neural networks (CNNs) for image classification, we aim to enhance the efficiency of glitch identification and removal. This will not only improve the quality of gravitational wave data but also aid in the continuous refinement of LIGO's detection capabilities. While this proposal is not new, our research focuses on multiview classification and explores various fusion strategies to optimize this process. By implementing these state-of-the-art models, we strive to contribute to the ongoing efforts to refine gravitational wave astronomy and deepen our understanding of the universe.

Introduction

Machine learning has been extensively utilized across various scientific domains, such as astronomy, biology, and climate science, to analyse complex data and enhance human efficiency. It has enabled researchers to make groundbreaking discoveries and advancements in their respective fields.

Gravitational-wave astronomy offers unique insights into our universe. The Advanced Laser Interferometric Gravitational-Wave Observatory (LIGO) and Advanced Virgo detectors have detected signals from interacting binaries of neutron stars and black holes.^[3] The rate of discovery has increased as the sensitivity of the detector network improves.^[4] The LIGO Scientific, Virgo, and KAGRA (LVK) Collaboration identified three candidates in the first observing run (O1), eight in the second observing run (O2), and seventy-nine in the third observing run (O3).^[5] The fourth observing run (O4) is ongoing, with more discoveries expected as the detectors continue to advance in sensitivity and accuracy.

These observations require measurements equivalent to fractional changes in distance on the order of 10^{-21} , necessitating careful isolation from noise. These noises, or glitches, pose significant challenges for gravitational-wave signal identification and analysis. There are many different glitch types, some with known environmental or instrumental origins and others with uncertain or unknown sources. Despite efforts to eliminate noise, it cannot be entirely eradicated, making it essential to understand the properties of noise in gravitational-wave detectors.

Identifying these glitches is crucial for improving gravitational-wave data quality. These categorized glitches can then be studied to identify their causes and mitigate them, either by eliminating their sources or minimizing their impact through adjustments in detection settings. In recent years, machine-learning methods have been developed for glitch analysis and classification in gravitational-wave detectors.^[6]

Compiling a catalogue of classified glitches is also useful for identifying their physical causes and evaluating their impact on data analysis. Previous classifications, for example, have been used to select examples of glitches for impact evaluation, study glitch morphology, cross-reference glitches with environmental noise or auxiliary-channel measurements, and train gravitational-wave detection and glitch classification algorithms.^[7] Identifying new classes of glitches can indicate new sources of noise and suggest areas for further investigation.

In this paper, we focus on multiview fusion. We discuss early, intermediate, and late fusion models that we have developed, studied, and assessed over the past two months. These fusion models have shown promising results in improving detection accuracy and robustness by combining information from multiple viewpoints. By leveraging the strengths of each fusion model, we aim to provide a comprehensive analysis of their performance in glitch detection systems.

By illuminating the advancements at the forefront of gravitational wave research, this paper endeavours to contribute to a deeper understanding of the cosmos and pave the way for future discoveries in this captivating field of study.

Proposed Model

Here, we present the development of a multiview deep neural network framework^[8, 9] for glitch classification. Compared to standard methods that use just one set of images, we propose four-input models. We exploit four different time durations that are available for each glitch, namely each glitch plotted over time windows of 0.5, 1, 2, and 4 seconds. Also, we have been working on three fusion strategies: early, intermediate, and late fusion. Previous experiments have already showed that multiview models outperform single view models.^[10] Here we have been trying to develop these models on our own and check their performance against each other in glitch classification. Our goal is to create a robust and accurate framework that can effectively classify glitches in distinct classes with high precision and efficiency.

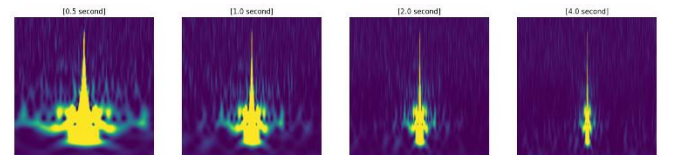


Fig 1: A Sample glitch in 4 Timeframes [0.5, 1.0, 2.0 and 4.0 seconds] as 4 Views.

Dataset

We primarily used the dataset available from the O3 run.^[11] However, we had to modify it in accordance with our needs and sort it out into different views. It consisted of a total of 22348 samples for training, 4800 for validation, and 4720 for testing, distributed over 22 classes and four views. Each view contains glitches in specific timeframes, i.e., 0.5 second, 1.0 second, 2.0 second, and 4.0 second.

We created two datasets of our own: one consisting of NumPy files (.npy) of image data stored in array format^[12], and the other simply contains images sorted into different view folders.^[13] Below we will discuss each model individually.

	Train	Test	Validation
1080 Lines	229 x 4	50 x 4	49 x 4
1400 Ripples	59 x 4	9 x 4	13 x 4
Air_Compressor	41 x 4	9 x 4	8 x 4
Blip	1274 x 4	273 x 4	275 x 4
Chrip	41 x 4	10 x 4	9 x 4
Extremely_Loud	316 x 4	64 x 4	67 x 4
Helix	195 x 4	42 x 4	42 x 4
Koi_Fish	498 x 4	102 x 4	106 x 4
Light_Modulation	361 x 4	78 x 4	73 x 4
Low_Frequency_Burst	437 x 4	90 x 4	94 x 4
Low_Frequency_Lines	351 x 4	66 x 4	66 x 4
No_Glitch	107 x 4	21 x 4	22 x 4
None_of_the_Above	57 x 4	11 x 4	13 x 4
Paired_Doves	19 x 4	4 x 4	4 x 4
Power_Lines	314 x 4	68 x 4	67 x 4
Repeating_Blips	185 x 4	37 x 4	41 x 4
Scattered_Light	308 x 4	67 x 4	68 x 4
Scratchy	237 x 4	50 x 4	50 x 4
Tomte	73 x 4	13 x 4	17 x 4
Violin_Mode	284 x 4	64 x 4	64 x 4
Wandering_Lines	29 x 4	7 x 4	6 x 4
Whistle	208 x 4	45 x 4	46 x 4

Table 1: Data Distribution over various glitch classes for Train, Test and Validation Datasets.

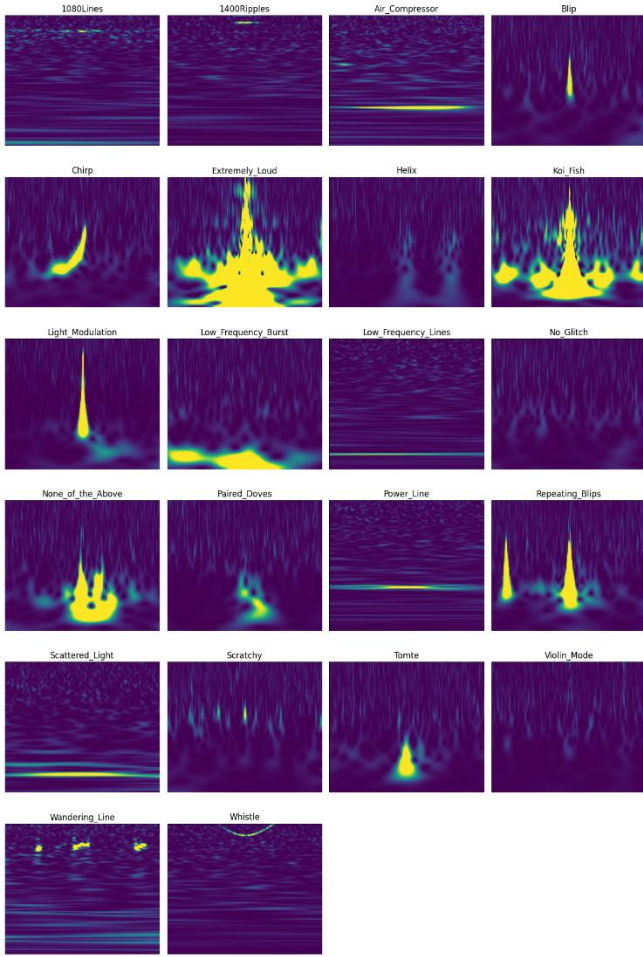


Fig 2: A sample from each 22 classes of glitches.

a. Early Fusion

The early fusion model^[14] was an advanced convolutional neural network (CNN) architecture designed to process multi-view images efficiently. Unlike traditional single-view CNNs, this model integrated multiple perspectives of an object at the initial stage, enhancing its ability to recognize and classify complex visual patterns.

Working

The model began by juxtaposing images from four different views in a 2x2 grid format. This configuration resulted in a single, larger image that effectively contained all perspectives of the object. By consolidating these views into one composite input, the model leveraged the combined visual information to make more informed predictions. The input dimensions were doubled compared to a single-view image, accommodating the four 224x224 images in one 448x448 image.

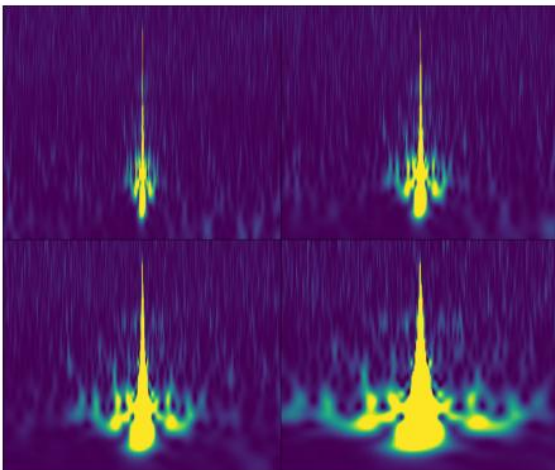


Fig 3: A sample of Concatenated Image used for Early Fusion Model.

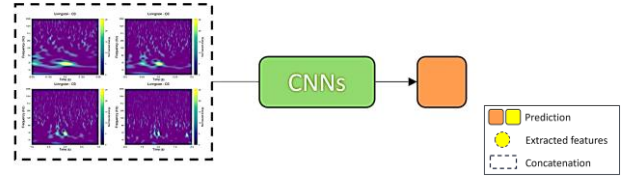


Fig 4: Model Architecture for Early Fusion Model*.

Data Handling and Preprocessing

Given the extensive computational resources required for handling multi-view images, the model utilized NumPy files for efficient data storage and retrieval. Custom-defined data generators were employed to manage memory usage effectively, especially important when working with limited RAM. These generators loaded data in batches, ensuring that the model was continuously fed with the necessary input without overwhelming the system's memory.

The preprocessing of images included several critical steps:

- Conversion to RGB Format: Ensured consistency in colour channels across all images.
- Rescaling: Normalized pixel values to enhance the model's performance.
- Resizing: Standardized image dimensions to 224x224 pixels for uniformity.

All preprocessing steps were completed before packing the data into NumPy files, streamlining the training process, and ensuring data consistency.

Training Process

Training the early fusion model involved shuffling the data to prevent overfitting and improve generalization. A batch size of 32 was used, balancing the computational load and training efficiency. During training, the custom-defined generator juxtaposed the four views on-the-go, feeding the composite images into the model.

To ensure optimal performance, model checkpoints were employed to save the best-performing model based on validation accuracy and loss.^[15] This approach allowed for robust model selection, facilitating the achievement of high accuracy and low loss metrics.

Performance Metrics

The early fusion model achieved impressive performance metrics:

Accuracy: 0.9585

Loss: 0.2377

These results indicated the model's high proficiency in classifying multi-view images, demonstrating its potential for applications requiring detailed visual analysis from multiple perspectives.

b. Late fusion

The late fusion model^[16] represented a sophisticated approach to multi-view image processing by integrating the probabilities from multiple models at the final stage to make decisions. This method allowed for the aggregation of insights from separate models, each trained on a different view, thus enhancing the overall predictive accuracy and robustness of the system.

Working

In the late fusion model, four separate CNNs were trained on the respective view data independently. After training, a custom 'LateFusion' class was defined to aggregate the predictions from these models. This class averaged the probabilities from all four models to make the final decision. By combining the outputs at the probability level, the model benefited from the strengths of each individual view's model, leading to more accurate and reliable predictions.

The 'LateFusion' class was a pivotal component of this model. It not only aggregated the predictions but also calculated the overall accuracy and loss of the combined model. Additionally, it generated detailed classification reports and plotted confusion matrices,

providing comprehensive insights into the model's performance. This class played a crucial role in managing the integration and evaluation of the multi-view data.

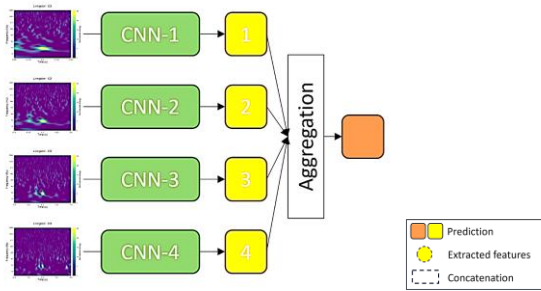


Fig 5: Model Architecture for Late Fusion*.

Data Handling and Preprocessing

For the late fusion model, a different dataset containing images was utilized. The Image Data Generator was employed for training, which handled data preprocessing dynamically during training. The preprocessing steps included:

- Conversion to RGB Format: Ensuring consistency in colour channels across all images.
- Rescaling: Normalizing pixel values to enhance the model's performance.
- Resizing: Standardizing image dimensions to 224x224 pixels for uniformity.

The use of the Image Data Generator allowed for efficient data handling and preprocessing on-the-fly, streamlining the training process.

Training Process

Training involved shuffling the data to prevent overfitting and ensure better generalization. A batch size of 32 was employed, balancing the computational load and training efficiency. The training process was managed by the Image Data Generator, which fed the data into the models in a continuous and efficient manner. A model checkpoint was employed to save the best-performing model based on validation metrics.

Performance Metrics

The late fusion model achieved impressive performance metrics:

Accuracy: 0.9576

Loss: 0.1960

These results indicated the model's high proficiency in classifying multi-view images, demonstrating its potential for applications requiring integrated insights from multiple perspectives.

c. Intermediate Fusion

The intermediate fusion model^[17] was another sophisticated convolutional neural network (CNN) architecture designed to enhance the processing of multi-view images. Unlike the early fusion model, which combined views at the input stage, the intermediate fusion model individually extracted features from each view before concatenating them. This method allowed for a more nuanced understanding of each perspective, which was then integrated to make more accurate predictions.

Working

In this model, a CNN separately processed each view to extract its features. These features were then concatenated, forming a comprehensive representation of the object from all perspectives. The concatenated features were weighted and fed into subsequent layers for decision-making. This process allowed the model to focus on the most relevant aspects of each view, improving its classification performance.

To further enhance the model's capabilities, custom attention layers and Incep-Res blocks^[18] were defined and utilized:

- Custom Attention Layer:** This layer helped the model focus on the most significant parts of the feature maps by assigning higher weights to the more important features. This selective focus improved the model's ability to discern subtle differences across views.
- Incep-Res Blocks:** These blocks combined the strengths of Inception modules and Residual networks, allowing the model to capture multi-scale features while maintaining efficient gradient flow. The Incep-Res blocks contributed to improved feature extraction and overall model robustness.

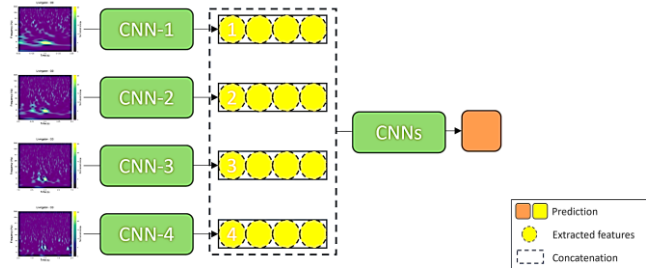


Fig 6: Model Architecture for Intermediate Fusion*.

Data Handling and Preprocessing

As with the early fusion model, the intermediate fusion model used the same NumPy dataset for data storage and retrieval. However, a different custom data generator was employed to manage data loading and conversion. This generator loaded data in batches from the NumPy files and converted them into tensors suitable for training.

Preprocessing steps included:

- Conversion to RGB Format: Ensuring consistency in colour channels across all images.
- Rescaling: Normalizing pixel values to improve the model's performance.
- Resizing: Standardizing image dimensions to 224x224 pixels for uniformity.

All preprocessing was performed before packing the data into NumPy files, maintaining consistency and efficiency in data handling.

Training Process

The training process for the intermediate fusion model involved data shuffling to prevent overfitting and ensure better generalization. A batch size of 32 was employed, balancing the computational load and training efficiency. The model checkpoint technique was utilized to save the best-performing model based on validation metrics.

Performance Metrics

The intermediate fusion model achieved notable performance metrics:

Accuracy: 0.9440

Loss: 0.4049

These results indicated the model's effectiveness in handling multi-view images, though with slightly lower accuracy and higher loss compared to the early fusion model.

Discussion

The proposed study explores the classification of gravitational wave glitches using advanced machine learning techniques, particularly focusing on multiview fusion strategies. Our work has demonstrated the effectiveness of early, intermediate, and late fusion models in handling the intricate task of glitch identification, which is crucial for the improvement of gravitational wave data quality and the overall efficiency of astrophysical explorations.

Performance Comparison and Analysis

Our experiments with the early, intermediate, and late fusion models reveal several noteworthy findings:

The early fusion model, which combined multiple perspectives at the initial stage, achieved the highest accuracy of 95.85%. This model's approach of creating a composite image from multiple views likely facilitated a more holistic analysis, allowing the CNN to capture complex visual patterns more effectively. However, the computational resources required for handling such large composite images were considerable, and this might present a limitation in real-world applications where computational efficiency is crucial. Future research could focus on optimizing the computational aspects of early fusion models, perhaps through more efficient data handling techniques or hardware advancements.

The intermediate fusion model, which extracted features from each view separately before concatenating them, also performed well, with an accuracy of 94.40%. This approach allowed for a more detailed extraction of features from each individual view, enhancing the model's ability to discern subtle differences across views. The incorporation of custom attention layers and Incep-Res blocks further improved feature extraction, suggesting that such architectural enhancements can significantly benefit multiview models. However, the slightly lower accuracy compared to the early fusion model indicates that, while intermediate fusion is effective, it may require further optimization to match the performance of early fusion. Note that creating this model took more time than other two, therefore we couldn't properly optimize it for the best performance. Future work could explore additional architectural improvements or feature engineering techniques to boost the intermediate fusion model's performance.

The late fusion model, which integrated predictions from separate models at the final stage, demonstrated a balanced approach with an accuracy of 95.76%. This method's strength lies in its ability to combine the strengths of individual view-specific models, thereby improving overall prediction reliability. The late fusion model's use of a custom class to aggregate predictions and generate comprehensive performance reports highlighted the practical advantages of this strategy in real-time glitch detection and analysis. The slightly lower accuracy compared to the early fusion model suggests that, while late fusion is robust and practical, it may not capture the intricacies of glitch patterns as effectively as early fusion. Future research could investigate hybrid fusion strategies that combine elements of early, intermediate, and late fusion to leverage the strengths of each approach.

Parameters	Early Fusion	Intermediate Fusion	Late Fusion
Filter Size	6 x 6	3 x 3	3 x 3
Batch Size	32	32	32
Input Image Size	448 x 448 x 3	224 x 224 x 3	224 x 224 x 3
Input Methodology	Custom Generator	Custom Generator	Keras Generator
Optimizer	Adam	Adam	Adam
Loss Function	Categorical Cross Entropy	Categorical Cross Entropy	Categorical Cross Entropy
Learning Rate	Default (0.001)	Default (0.001)	Default (0.001)
Callbacks	Model Checkpoints	Model Checkpoints	Model Checkpoints
Total Trainable Parameters	35,642,614 (135.97 MB)	12,015,126 (2.65 GB)	9,689,814 x 4 (36.96 MB x 4)
Complexity	Basic	Complex	Basic
Accuracy	0.9585	0.9440	0.9576
Loss	0.2377	0.4049	0.1960
Potential of Improvement	Average	High	Minimal

Table 2: Parameters Comparison and Analysis Table

Overall, the findings underscore the importance of multiview approaches in enhancing the classification of glitches in gravitational wave data. Each fusion strategy has its own unique advantages, and the choice of model may depend on specific application requirements, such as the need for computational efficiency, real-time processing capabilities, or the highest possible accuracy. Future research could focus on further optimizing these models, exploring

hybrid fusion strategies that combine the strengths of early, intermediate, and late fusion, and applying these techniques to larger and more diverse datasets to validate their scalability and generalizability.

Working Approach

Our research journey involved the individual development of three fusion models: early fusion, intermediate fusion, and late fusion. A significant common challenge we faced was adapting the already available dataset for use in multiview models, which required data sorted into multiple views.

To address this, we initially decided to create a separate dataset tailored to our research needs. We reorganized the existing data into four distinct views to facilitate multiview analysis. However, training the models directly on this new dataset presented another challenge. Limited RAM posed a significant problem, as loading the entire dataset at once was not feasible. This led us to explore the concept of data generators.

Data generators are mechanisms that load data dynamically during training, thus reducing the memory load. This approach worked well for the late fusion model, which benefited from the on-the-fly data loading capabilities of Keras built-in data generators. However, the early and intermediate fusion models required more complex input handling—concatenated images for early fusion and tensor data for intermediate fusion—which could not be managed by standard data generators.

To overcome this, we experimented with converting image data into NumPy files, aiming to streamline data loading. Unfortunately, this approach still resulted in high RAM usage, negating the intended benefits. After further exploration, we decided to define custom data generators tailored specifically to the early and intermediate fusion models. These custom generators could handle the unique requirements of concatenated images and tensor data, respectively, ensuring efficient data feeding without overwhelming system resources.

Despite solving the data feeding issue, we encountered additional challenges related to the compatibility of custom-defined classes with the new model architectures and generators. These classes had to be modified and serialized to adapt to the new setup, which required careful debugging and optimization.

The late fusion model was completed first, allowing ample time for optimization. Similarly, the early fusion model was ready relatively quickly, enabling us to implement various enhancements. However, the intermediate fusion model faced more complex issues, resulting in a longer development time and less opportunity for optimization.

In conclusion, while we achieved functional models for all three fusion strategies, there remains significant scope for further optimization and performance improvement. Our research laid a solid foundation, but future work could enhance these models' efficiency and accuracy, making them even more robust for glitch classification in gravitational wave data. By addressing the remaining challenges and refining our approach, we believe these models have the potential to perform even better, contributing valuable advancements to the field of gravitational wave astronomy.

Comparative Advantages of Fusion Strategies

Each fusion strategy brings distinct advantages to the table:

Early Fusion: Simplifies the integration process by consolidating views early on, which can be computationally efficient and effective for models capable of handling large input dimensions. It leverages the full spectrum of visual data from the onset, making it ideal for scenarios where comprehensive visual representation is beneficial.

Intermediate Fusion: Offers a balanced approach by allowing individual feature extraction before combining the views. This strategy is particularly advantageous in capturing detailed and nuanced patterns from each view, making it suitable for complex classification tasks where different views provide complementary information.

Late Fusion: Excels in integrating insights from independently trained models, making it highly effective in scenarios where different views might offer conflicting information. This approach allows for more flexible and potentially more accurate decision-making by aggregating final predictions.

Conclusion

Practical Implications and Future Work

The successful implementation of these fusion models has significant practical implications for the field of gravitational wave astronomy. Enhanced glitch classification can lead to improved data quality, enabling more accurate detection of gravitational waves and deeper insights into astrophysical phenomena. This, in turn, supports the continuous refinement of detector sensitivity and the advancement of our understanding of the universe.^[19, 20]

Future research can expand on this work by exploring several avenues:

Further Optimization: Fine-tuning the hyperparameters and architectures of the fusion models could yield even better performance. Techniques such as automated machine learning (AutoML) could be employed to systematically explore a broader range of configurations.

Real-time Application: Implementing these models in real-time glitch classification systems could provide immediate benefits for ongoing gravitational wave observations. This would require efficient deployment strategies, potentially leveraging edge computing and parallel processing capabilities.

Transfer Learning: Applying transfer learning techniques could enhance the models' generalizability across different datasets and observational runs. By pre-training on large, diverse datasets and fine-tuning on specific glitch data, the models' robustness and accuracy can be further improved.

Cross-disciplinary Applications: The multiview fusion strategies developed in this study could be applied to other domains requiring multi-perspective data analysis, such as medical imaging, autonomous driving, and remote sensing. This cross-disciplinary application could foster innovations and improvements in various fields.

External Insights

Recent advancements in machine learning, particularly deep learning, have significantly impacted various scientific fields, including gravitational wave detection. Studies have shown that CNNs, especially when combined with advanced techniques like transfer learning and data augmentation, can dramatically improve the performance of image classification tasks. For instance, research by George and Huerta (2018) demonstrated that deep learning models could outperform traditional methods in identifying gravitational wave signals amid noise.

Moreover, the increasing availability of computational resources and large datasets has enabled more sophisticated models to be trained and deployed effectively. The success of multiview fusion in our study aligns with trends observed in other domains, where combining multiple perspectives often leads to superior performance compared to single-view approaches.

In conclusion, our study underscores the potential of multiview fusion strategies in enhancing glitch classification for gravitational wave detectors. The promising results achieved by the early, intermediate, and late fusion models highlight their respective strengths and pave the way for future research and practical implementations aimed at advancing gravitational wave astronomy and related fields.

References and Citations

[1] Einstein, A. (1916). The foundation of the general theory of relativity. *Annalen der Physik*, 354(7), 769-822.

[2] Abbott, B. P., et al. (2016). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116(6), 061102.

[3] Abbott, B. P., et al. (2019). GWTC-1: A gravitational-wave transient catalog of compact binary mergers observed by LIGO and Virgo during the first and second observing runs. *Physical Review X*, 9(3), 031040.

[4] Abbott, R., et al. (2021). GWTC-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run. *arXiv preprint arXiv:2111.03606*.

[5] J Glanzer, S Banagiri, S B Coughlin, S Soni, M Zevin, C P L Berry, O Patane, S Bahaadini, N Rohani1, K Crowston, V Kalogera, C Østerlund, L Trouille and A Katsaggelos. Data quality up to the third observing run of Advanced LIGO: Gravity Spy glitch classifications. *arXiv:2208.12849v2 [gr-qc]* 8 Feb 2023

[6] George, D., & Huerta, E. A. (2018). Deep neural networks to enable real-time multimessenger astrophysics. *Physical Review D*, 97(4), 044039.

[7] Zevin, M., et al. (2017). Gravity Spy: Integrating advanced LIGO detector characterization, machine learning, and citizen science. *Classical and Quantum Gravity*, 34(6), 064003.

[8] Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems*, 25, 1097-1105.

[9] Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.

[10] Sara Bahaadini, Neda Rohani, Scott Coughlin, Michael Zevin, Vicky Kalogera, Aggelos K Katsaggelos. Deep Multi-View Models for Glitch Classification. *arXiv:1705.00034v1 [cs.LG]* 28 Apr 2017

[11] Link to the Original Dataset:
<https://www.kaggle.com/datasets/tentotheminus9/gravity-spy-gravitational-waves>

[12] Link to the dataset consisting NumPy Files:
<https://www.kaggle.com/datasets/urnishapaul/gr4v17y-5py-dataset-multi-viewnumpy-format>

[13] Link to the dataset consisting of Raw Images:
<https://www.kaggle.com/datasets/n1kulx0/gr4v17y-5py-dataset-multi-view>

[14] Link to Early Fusion Code Notebook:
<https://www.kaggle.com/code/n1kulx0/gr4v17y-5py-earlyfusion-model>

[15] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.

[16] Link to Late Fusion Code Notebook:
<https://www.kaggle.com/code/n1kulx0/gr4v17y-5py-latefusion-model>

[17] Link to Intermediate Fusion Code Notebook:
<https://www.kaggle.com/code/n1kulx0/gr4v17y-5py-intermediatefusion-model>

[18] Yunan Wua, Michael Zevinbe, Christopher P. L. Berryc, Kevin Crowstond, Carsten Østerlundd, Zoheyr Doctore, Sharan Banagirie, Corey B. Jacksonf, Vicky Kalogerae, Aggelos K. Katsaggelosa. Advancing Glitch Classification in Gravity Spy: Multi-view Fusion with Attention-based Machine Learning for Advanced LIGO's Fourth Observing Run. *arXiv:2401.12913v1 [gr-qc]* 23 Jan 2024

[19] Dreissigacker, C., et al. (2019). Deep transfer learning for gravitational wave searches. *Physical Review D*, 100(4), 044009.

[20] The LIGO Scientific Collaboration, the Virgo Collaboration. (2020). GW190814: Gravitational waves from the coalescence of a 23 solar mass black hole with a 2.6 solar mass compact object. *The Astrophysical Journal Letters*, 896(2), L44.

*Images of Model Architecture were taken from other sources, [18] to be exact.

Project Timeline

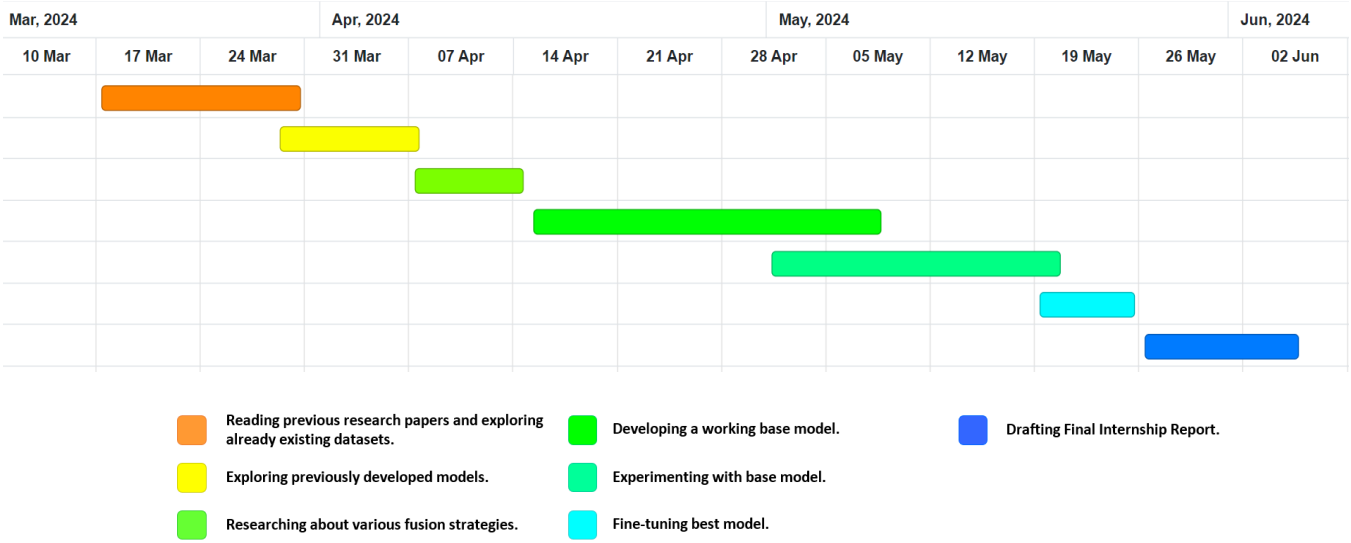


Fig 7: Gantt Chart.